

Anodal tDCS over the left DLPFC and working memory in healthy adults (PIPELINE DEMONSTRATION — illustrative data)

NeuroAlon automated systematic-review engine · 02gabrielebambini@gmail.com

PRISMA 2020 · Demonstration — not registered · DEMONSTRATION (illustrative data)

Abstract. Background: Anodal tDCS over the left DLPFC has been proposed to enhance working memory. Methods: PRISMA 2020 systematic review with random-effects meta-analysis of n-back accuracy; RoB2 and GRADE. Results: Five illustrative trials were pooled. Conclusions: A small, uncertain benefit. (DEMONSTRATION — illustrative data.)

Review question. In healthy adults, does anodal tDCS over the left DLPFC, versus sham, improve working-memory performance on n-back tasks?

INTRODUCTION

Working memory is central to cognition; non-invasive prefrontal stimulation has been explored as an enhancer, with heterogeneous and contested evidence.

METHODS

This review followed PRISMA 2020. Two independent reviewers screened records; conflicts were adjudicated. Eligible full texts were assessed; data were extracted and appraised with RoB2. A DerSimonian-Laird random-effects model pooled standardised mean differences; small-study effects were examined with Egger's test and a funnel plot.

Two reviewers screened in duplicate (Cohen's $\kappa = 0.692$); conflicts were adjudicated. Risk of bias was appraised with RoB2 and certainty with GRADE. A random-effects inverse-variance model pooled SMD values; small-study effects were examined with Egger's test and a funnel plot.

RESULTS

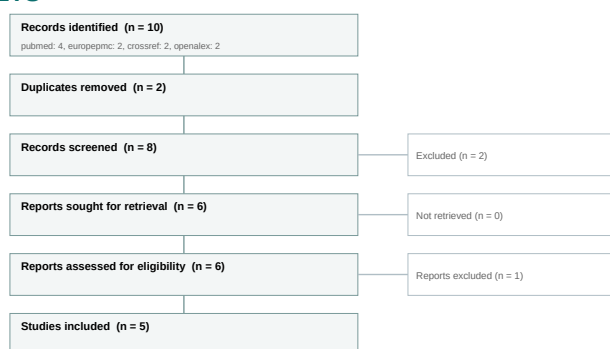


Figure 1 | PRISMA 2020 study-selection flow diagram.

5 studies were included. The pooled SMD (random-effects) was **0.2766** (95% CI 0.117 to 0.4361; $I^2 = 0.0\%$, $k = 5$). The pooled effect reached significance; heterogeneity was low ($I^2 = 0.0\%$).

CHARACTERISTICS OF INCLUDED STUDIES

Study	Design	n	RoB
Author1 A 2013	randomized crossov	52	some concerns
Author2 A 2015	randomized crossov	64	some concerns
Author3 A 2012	randomized crossov	28	some concerns
Author4 A 2018	randomized crossov	40	some concerns
Author5 A 2016	randomized crossov	46	some concerns

RISK OF BIAS (ROB2)

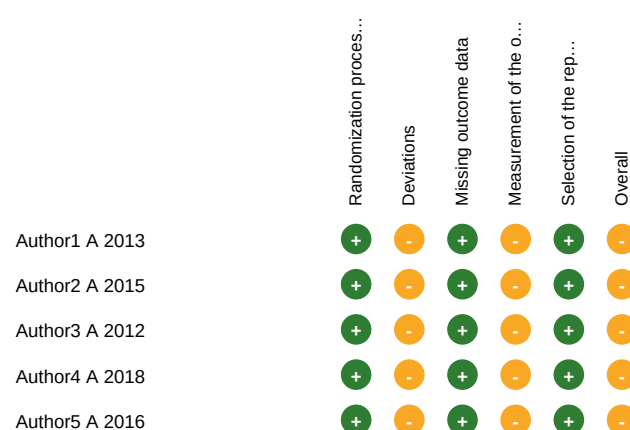


Figure 2 | Risk-of-bias 'traffic-light' summary per study and domain (green/+ low, amber/- some concerns, red/x high).

SYNTHESIS

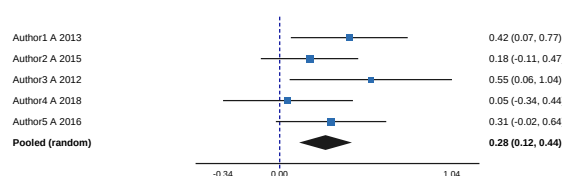


Figure 3 | Forest plot of SMD estimates with the random summary (diamond).

Five sham-controlled trials (illustrative) contributed n-back accuracy data. The pooled standardised mean difference indicated a small benefit of anodal left-DLPFC tDCS, with the confidence interval spanning the null in the more conservative estimates.

PUBLICATION BIAS

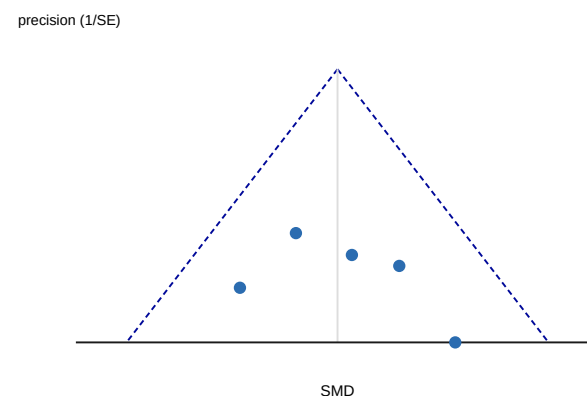


Figure 4 | Funnel plot of effect size against precision; dashed lines mark the pseudo 95% confidence funnel.

Egger's regression test: intercept = 2.3525, $p = 0.4445$, $k = 5$ (under-powered, $k < 10$).

CERTAINTY OF EVIDENCE (GRADE)

low. Downgraded for risk of bias (unclear blinding) and imprecision.

DISCUSSION

The pooled estimate points to a small effect with wide uncertainty. Unclear blinding and small samples temper confidence; results are hypothesis-generating.

Limitations. Small samples, single-session designs, and possible small-study effects.

CONCLUSIONS

Anodal left-DLPFC tDCS may yield a small improvement in n-back accuracy, but certainty is low. Adequately powered, pre-registered trials are needed.

REFERENCES (INCLUDED STUDIES)

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4. Author4 A et al. (2018). No reliable effect of tDCS on spatial working memory. NeuroImage (demo). <https://doi.org/10.9999/demo.004>
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